Chapter 19

Interpreting Pharmacoeconomic Findings

Piyameth Dilokthornsakul[1], Dixon Thomas[2], Lawrence Brown[3] and Nathorn Chaiyakunapruk[4]
[1]Naresuan University, Phitsanulok, Thailand; [2]Gulf Medical University, Ajman, United Arab Emirates; [3]Chapman University School of Pharmacy, Irvine, CA, United States; [4]Monash University Malaysia, Selangor, Malaysia

Learning Objectives:

Objective 19.1 Describe the influence of pharmacoeconomics in healthcare.
Objective 19.2 Explain how to interpret pharmacoeconomic analysis findings.

OBJECTIVE 19.1. DESCRIBE THE INFLUENCE OF PHARMACOECONOMICS IN HEALTHCARE

Pharmacoeconomics is the scientific discipline that identifies, measures, and compares the value and affordability of pharmaceutical products in the health system and society. *Pharmacoeconomics is commonly defined as the scientific discipline that evaluates the value of pharmaceutical products, services, or programs under individual, firms, government, or societal point of view*.[1] It is a part of health economics and more broadly a part of health technology assessment. It estimates incremental value of pharmaceutical interventions. The term "value" in general refers to the outputs achieved relative to the costs incurred when new is compared with old intervention(s). In healthcare, *value can be defined as the patient health outcomes achieved per money spent*.[2] This is one of the goals of pharmacoeconomics that addresses whether an additional gain from a medical intervention is worth paying extra cost based on three aspects of health outcomes: economic, clinical, and humanistic outcomes (known as the ECHO model).

Selection of drugs, whether included in a national or hospital formulary or consumed by a patient, is not an easy task. Cost and consequences should be considered when evaluating the value of new healthcare interventions compared to old ones. New healthcare interventions, including medical products or pharmacy-related services, are not always safer, effective, and affordable. Commonly, new medical interventions may be safer and/or more effective, but more expensive than existing alternatives. Some interventions can be very expensive causing financial burden to consumers and even the whole healthcare system. Therefore, resources should be wisely spent for sustainability.

Important questions are "Is the drug worth paying for?" and "Is the drug affordable within limited budgets?" To answer these questions, decision-makers need to assess the clinical benefits and costs of newer and existing drugs. It is obvious that clinical benefits and costs of drugs should be assessed in an organized way rather than using an informal assessment, such as "educated guess" or "what we did last time." Pharmacoeconomics provides systematic approaches to assess the clinical benefit and cost of drugs in a comparative manner.[3]

For example, an imaginary expensive lipid-lowering drug (Lipid-remove) was approved in a country. Data from clinical trials show its higher efficacy in lowering low-density lipoprotein cholesterol (LDL-c) than existing lipid-lowering drugs. However, the drug had a higher risk of myopathy and higher cost. The costs and outcomes of Lipid-remove and related hypothetical agents are given in Table 19.1.

Based on the information provided, an important question for health insurance programs is whether Lipid-remove should be included in their benefits packages? Some health insurance companies might decide to add Lipid-remove to their benefits packages but some might not. Without pharmacoeconomic evaluations, the decision might be based on only efficacy data (LDL-c reduction), only safety profile (risk of myopathy), or only cost based on their experience. Those strategies might not be appropriate for decision making because all important information is not appropriately considered collectively.

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00019-2
Copyright © 2019 Elsevier Inc. All rights reserved.

Pharmacoeconomics could help decision-makers to systematically consider the efficacy/effectiveness data (LDL-c reduction), safety profile (risk of myopathy), and cost. It could summarize those data into a single measure to inform decision-makers. This will help decision-makers follow a transparent and appropriate decision-making process.

Professional societies, governments, and universities are promoting best use of pharmacoeconomics. Good practice guidelines are available on different topics in pharmacoeconomics. Country-specific pharmacoeconomic guidelines are also available on the International Society for Pharmacoeconomics and Outcomes Research (ISPOR) website (www.ispor.org).[3]

Pharmacoeconomic analyses might be required when more than one healthcare intervention is available that varies in costs and/or outcomes. It is also required if only one expensive intervention is available, as its financial burden (cost of illness) and the budget impact need to be studied.

In addition to the comparative clinical benefits, pharmacoeconomic analyses provide more information that helps in the decision-making process. An analysis might not be needed for an intervention that is safer, more effective, and cheaper than its alternatives. However, when an intervention is not only more effective but also more costly to the comparator, making a decision needs more information about its value and affordability.

TABLE 19.1 A Hypothetical Example of Cost and Consequences

Medication	Status	LDL-c Reduction (%)	Risk of Myopathy (%)	Cost in USD (per year)
Lipid-remove	Newcomer	50	0.5	100,000
Lower-lipid-A	Existing one	40	0.2	80,000
Lipid-reduced-B	Existing one	45	0.1	60,000

LDL-c, low-density lipoprotein cholesterol; *USD,* US dollar.

According to the hypothetical example shown in Table 19.1, if a health insurance company would like to select either Lower-lipid-A or Lipid-reduced-B for their benefit package, the company does not need to perform pharmacoeconomic analysis because evidence shows that Lipid-reduced-B had a higher efficacy than Lower-lipid-A (45% vs. 40%), lower risk of myopathy (0.1% vs. 0.2%), and lower cost ($60,000 vs. $80,000). It is obvious that Lipid-reduced-B is more cost-effective than Lower-lipid-A.

In general, pharmacoeconomic analysis is needed when a medication of interest, usually a new medication, not only has a higher benefit than a comparator(s) but also has a higher cost or worse safety profile.

Application of pharmacoeconomics ranges from a specific cost-effectiveness of a group of drugs in clinical practice (microeconomics) to a wider cost-effectiveness of a group of public health programs (macroeconomics).[4] As an example, it can range from micro- to macroeconomics as follows:

- cost-minimization in a group of patients;
- cost-effectiveness and budget impact for a health insurer; and
- pricing of pharmaceuticals or resource allocation in a national drug budget.[5]

However, the most common application of pharmacoeconomics is in health insurance reimbursement decisions.

Many countries use pharmacoeconomics as an analytic tool in health technology assessment for new interventions. Australia was the first country to publish a guideline to perform pharmacoeconomic analysis in 1992, and they require pharmaceutical companies to submit pharmacoeconomic data for pharmaceuticals that the companies would like added to the national Pharmaceutical Benefits Scheme (PBS).[6,7] The Pharmaceutical Benefits Advisory Committee (PBAC) in Australia recommends new medications for listing on the PBS. PBAC uses information on comparative clinical effectiveness, safety, and pharmacoeconomics of new medications to be considered for recommendations.[8]

Soon after the publication of the Australian pharmacoeconomic guideline, Canada initiated a health technology assessment agency called "the Canadian Agency for Drugs and Technologies in Health (CADTH)."[7] CADTH has a program called "CADTH common drug review (CDR)," which aims to deliver evidence and reimbursement recommendations to Canada's federal, provincial, and territorial public drug plans to guide their drug funding decisions.[9]

Another well-known health technology appraisal agency that uses pharmacoeconomics as important information for making reimbursement recommendations is the National Institute for Health and Care Excellence (NICE) of the United Kingdom.[10] It was initiated in 1999 with the aim to provide national guidance and advice to improve health and social care. NICE has several centers to develop guidance for improving health and social care, including the Centre for Health

Technology Evaluation, which is responsible for making recommendations for reimbursement using clinical and economic evidence appraisal.

In the United States, there are many health technology assessment agencies that include pharmacoeconomics as one of their works. However, the government agency that is responsible for conducting health technology assessment is "Agency for Healthcare Research and Quality (AHRQ)."[11] AHRQ's mission is to produce evidence to make healthcare safer, higher quality, more accessible, equitable, and affordable. It provides technology assessments for the Centers for Medicare and Medicaid Services to inform its national coverage decision for the Medicare program.

Many other countries have health technology assessment agencies: Sweden, the Netherlands, New Zealand, Belgium, France, Germany, Italy, Spain, Taiwan, Thailand, South Korea, etc.[12–16]

OBJECTIVE 19.2. EXPLAIN HOW TO INTERPRET PHARMACOECONOMIC ANALYSIS FINDINGS

Pharmacoeconomic analyses typically measure both costs and consequences of treatments. There are related studies that measure only costs or only consequences. From Table 19.2, it is clear that full pharmacoeconomic evaluations compare both cost and consequences of at least two interventions.[17]

- Three types of descriptive studies describe certain features of an intervention, but do not compare it with another intervention. An *outcome description study* examines only health outcomes (i.e., blood pressure, hemoglobin A1C, fasting blood sugar, the number of clinical events, quality of life, etc.), and it does not compare an intervention to others. A *cost description study* (e.g., cost of illness describing the costs of an illness) examines only cost with no comparison of an intervention to others. A *cost outcome description study* examines both cost and health outcomes with no comparison of an intervention to others.
- A study that compares only health outcomes of an intervention to other intervention(s) is called an *efficacy or effectiveness study.*
- A study that compares only costs of an intervention to other intervention(s) is called a *cost analysis study.* An example is a budget impact analysis.
- A study that compares both cost and consequences of an intervention to other intervention(s) is classified as *full economic evaluation.* Examples are cost-effectiveness analysis (CEA), cost-benefit analysis (CBA), cost-utility analysis (CUA), cost-minimization analysis (CMA), etc.

For example, a study was conducted to determine hospitalization cost for elective percutaneous coronary intervention in five public cardiac centers using a cross-sectional study.[18] The authors reported that the average hospitalization cost of elective percutaneous coronary intervention ranged from $3186 to $4018. This study can be classified as a *cost description study* because the study examined only hospitalization cost for elective percutaneous coronary intervention with no comparison to other interventions and no examination of health outcomes.

Another study was conducted to determine cost, life years, and quality-adjusted life years (QALYs) of percutaneous coronary intervention with drug-eluting stents compared to bypass surgery for patients with three vessels or left main coronary artery disease.[19] The authors reported that cumulative life years for patients receiving percutaneous coronary intervention and bypass surgery were 4.601 and 4.701 for 5 years of observation, respectively, whereas the cumulative costs of those interventions were $47,641 and $53,260, respectively. This study can be classified as a *full economic evaluation* because it compared both costs and health outcomes for at least two interventions.

TABLE 19.2 Types of Studies Based on the Examination of Costs and Consequences

Categories		Costs and Consequences Examined		
		Health Outcomes Only	Cost Only	Both Health Outcomes and Cost
Comparison (at least 2 interventions)	No	Outcome description	Cost description	Cost outcome description
	Yes	Efficacy/effectiveness evaluation	Cost analysis	Full economic evaluation

Adapted from Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Healthcare Programmes.* 4th ed. Oxford, United Kingdom: Oxford University Press; 2015.

There are at least four types of pharmacoeconomic analyses[17,20]:

- CEA
- CUA
- CMA
- CBA

TABLE 19.3 Types of Pharmacoeconomic Analysis

Type of Pharmacoeconomic Analysis	Cost	Outcomes
Cost-minimization analysis	Monetary value	–
Cost-effectiveness analysis	Monetary value	Health outcomes in a natural unit (i.e., hemoglobin A1C, low-density lipoprotein cholesterol)
Cost-utility analysis	Monetary value	Outcome in a common unit (i.e., quality-adjusted life year)
Cost-benefit analysis	Monetary value	Monetary value

The main difference among these analyses is the unit of health outcomes measured and its implications. Many professionals consider that CUA and CMA are different types of CEA. The CMA assumes that the outcomes are identical for an intervention or its comparator, so if there is a difference in cost, the cheapest intervention could be adopted. CEA is the prototype pharmacoeconomic evaluation that measures the outcomes in natural units, e.g., hemoglobin A1C levels for hypoglycemic agents, LDL-c levels for lipid-lowering agents, life years saved for any intervention affecting mortality rate, etc. The CUA measures health outcomes in a universal unit like the QALY or disability-adjusted life years. Even though technically QALY is used in CUA, some professionals use it as an outcome measure but call the study CEA. The CBA is quite different in the sense that the health outcomes are converted into a currency unit, which is clearly an unnatural unit and sometimes unethical (Table 19.3).

For example, a study was conducted to evaluate long-term cost-effectiveness of insulin detemir compared with insulin glargine in patients with type II diabetes.[21] The authors reported that insulin detemir yielded greater QALYs with higher lifetime cost than insulin glargine. They also reported that health agencies might need to pay 1.7 million US dollars for one additional QALY gained. The study is considered as CUA (some professionals might call it CEA, which is acceptable in common practice) because it measures QALY as the outcome of interest, which is the outcome in a common unit. In fact, when the outcomes are measured in QALY in real practice, the study could be called CUA or CEA. Differentiating CUA with CEA using QALY is only technical and is not observed in real practice.

Because pharmacoeconomic analyses estimate both cost and health outcome and compare such cost and outcome of one intervention with others, there is a specific way to report their findings. In general, pharmacoeconomic analysts report their findings in one common measure "Incremental cost-effectiveness ratio (ICER)."[17,20,22] ICER is calculated by the difference in cost between an intervention of interest and a comparator divided by the difference in health outcomes between the intervention of interest and the comparator (Eq. 19.1). CEA and CUA use the same formula for ICER. When QALY is used as the outcome measure, it could be called CUA or CEA. When natural units such as A1C or LDL-c are used as an outcome measure, it is called CEA.

$$\text{ICER} = \frac{\text{Total cost}_{\text{intervention}} - \text{Total cost}_{\text{comparator}}}{\text{Outcomes}_{\text{intervention}} - \text{Outcomes}_{\text{comparator}}} \tag{19.1}$$

Suppose a medication called "SugarLower" comes on the market as a new glucose-lowering agent, whereas another medication called "GlucoseReduction" is the most effective glucose-lowering agent currently available in the market for patients with type II diabetes mellitus. Estimated total lifetime cost and QALYs for both medications are shown in Table 19.4.

TABLE 19.4 A Hypothetical Example of the Incremental Cost-Effectiveness Ratio

Medication	Status	QALYs	Total Cost per Lifetime (USD)
SugarLower	Newcomer	40	15,000,000
GlucoseReduction	Existing one	38	14,500,000

QALYs, quality-adjusted life years; *USD,* US dollar.

An example of CUA analyzing the value of SugarLower compared with GlucoseReduction. An ICER of the analysis is calculated as shown below:

$$\text{ICER} = \frac{15{,}000{,}000 - 14{,}500{,}000}{40 - 38} = 250{,}000 \text{ USD/QALY}$$

It could be interpreted that health agencies need to spend an additional 250,000 US dollars for SugarLower to gain one additional QALY compared with GlucoseReduction for patients with type II diabetes mellitus.

An ICER can provide valuable information on how much health agencies need to spend for additional health benefit, but it cannot determine whether the new intervention is worth purchasing. Worthiness is dependent on each context. For example, if a health insurance company (company A) has a willingness-to-pay threshold of 100,000 US dollars for an additional QALY, they might decide not to add SugarLower to their benefits package. On the other hand, another health insurance company (company B) would like to cover SugarLower for their health beneficiaries as the company has a higher willingness-to-pay threshold of 300,000 US dollars per one additional QALY gained. Thus, company B might add SugarLower to their benefits package, while company A would likely not.

An important issue for ICER is that a new intervention does not always come with higher total costs and better health outcomes. Some new interventions come with a lower total cost and a better health outcome and some come with a higher total cost but a worse health outcome. Thus, the ICER should be interpreted correctly in such situations. Therefore, a graphical presentation of ICER findings, which is known as a "cost-effectiveness analysis plane (CEA plane)" could help in the easy interpretation of the results (Fig. 19.1).

A CEA plane provides the relationship between comparative cost and health outcome. The X-axis indicates incremental health outcome (also called incremental effectiveness), whereas the Y-axis indicates incremental cost. The CEA plane is divided into four quadrants. Quadrant 1 has better health outcomes and a higher total cost of a new intervention compared to a comparator. An ICER is needed to be calculated in this quadrant to determine how much health agencies need to spend for additional health outcome gained. The ICER threshold or willingness-to-pay threshold by the health insurance company needs to be compared to the calculated ICER. Most CEAs fall into quadrant 1.

Quadrant 2 stands for worse health outcomes and higher total cost of a new intervention. The intervention that falls in quadrant 2 should not be considered for coverage because it loses some health outcome and increases the total cost of the treatment. The intervention in quadrant 2 is called "dominated intervention" and therefore is a clear reject.

Quadrant 3 stands for worse health outcomes but a lower total cost of a new intervention. The intervention that falls in quadrant 3 is debatable. Some healthcare providers would not consider the intervention in this quadrant because it provides worse health outcomes. Thus, cost should not be a factor to be considered. On the other hand, some healthcare providers consider the intervention in this quadrant because it is the bargain between worse health outcomes and a lower total cost. Other factors should be considered for the intervention in this quadrant.

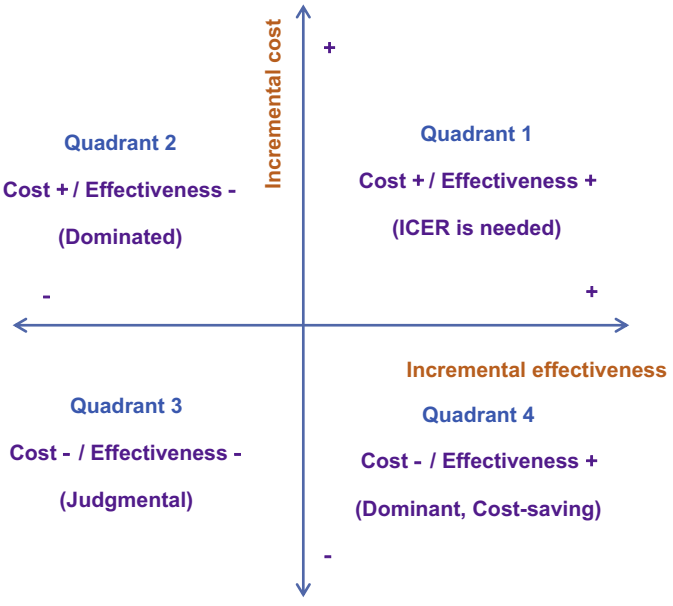


FIGURE 19.1 Cost-effectiveness analysis plane.

TABLE 19.5 ICER of Saxagliptin and a Sulfonylurea in a Study From Thailand

Medication	QALYs	Total Cost (US Dollars)	ICER (US Dollars/QALY)
Saxagliptin	7.552	17,316	76,140
Sulfonylurea	7.528	15,474	

ICER, incremental cost-effectiveness ratio; *QALYs,* quality-adjusted life years; *US,* the United States.

Quadrant 4 stands for better health outcomes and a lower total cost of a new intervention. The intervention that falls in quadrant 4 should be considered for health coverage because it gains some health outcome and decreases the total cost of treatment. The intervention in quadrant 4 is called "dominant intervention" and therefore is a clear acceptance.

According to the hypothetical example in Table 19.4, SugarLower has better QALYs (40 vs. 38 QALYs) and higher cost (15,000,000 vs. 14,500,000 US dollars). SugarLower falls into quadrant 1. Thus, an ICER is needed to determine the value of SugarLower compared with GlucoseReduction.

For instance, a CUA was conducted to determine the value of dipeptidyl peptidase-4 inhibitor monotherapy compared with sulfonylurea monotherapy for type 2 diabetic patients with chronic kidney disease in a study from Thailand.[23] The findings are presented in Table 19.5.

The study indicated that the lifetime cost of saxagliptin was $17,316 with 7.552 QALYs, whereas the cost of sulfonylurea was $15,474 with 7.528 QALYs. Saxagliptin is in quadrant 1 (higher cost with better health benefits). Therefore, an ICER was needed to determine the value of saxagliptin compared with sulfonylurea. The difference in cost was calculated as $17,316 − $15,474 = $1842, and the difference in health benefit was calculated as 7.552 − 7.528 = 0.024 QALY. The ICER was $76,140 per QALY. The willingness-to-pay threshold in the country (Thailand) was 4435 US dollars per QALY gained. The CEA finding showed that saxagliptin was not a cost-effective option because the ICER was higher than the willingness-to-pay.

Another example is a CUA of intravenous immunoglobulin for the treatment of steroid-refractory dermatomyositis in a study from Thailand.[24] It compared the cost and QALYs of intravenous immunoglobulin plus corticosteroid and immunosuppressant plus corticosteroid. The findings are presented in Table 19.6.

The findings indicated that intravenous immunoglobulin plus corticosteroids had better health outcome. Incremental QALY was 1.964. The intervention also had a lower total cost. The incremental cost was $4738. Thus, intravenous immunoglobulin plus corticosteroids fell in quadrant 4 (better health outcomes and lower cost). Thus, there was no need to calculate an ICER. The findings showed that intravenous immunoglobulin plus corticosteroids was the dominant intervention compared with immunosuppressant plus corticosteroid in patients with steroid-refractory dermatomyositis in Thailand.

Pharmacoeconomic Reporting Guidelines

Reporting of pharmacoeconomic evaluation is very important, but often poorly done and this limits the use of pharmacoeconomics by decision-makers. Different guidelines are available for conducting and reporting economic evaluations. One of these is a Consolidated Health Economic Evaluation Reporting Standards (CHEERS) statement, which provides recommendations in a checklist (Table 19.7) to optimize reporting of economic evaluations. As per the CHEERS

TABLE 19.6 ICER for Two Treatment Options in Steroid-Refractory Dermatomyositis in a Study From Thailand

Medication	QALYs	Total Cost (US Dollars)	Incremental QALY	Incremental Cost
Intravenous immunoglobulin plus corticosteroids	6.297	45,385	1.964	−4738
Immunosuppressant plus corticosteroids	4.33	50,124		

ICER, incremental cost-effectiveness ratio; *QALYs,* quality-adjusted life years; *US,* the United States.

TABLE 19.7 Items Present in a Typical Pharmacoeconomic Analysis as per CHEERS Statement 2013[25]

Section	Item No.	Recommendation to Authors of Pharmacoeconomic Studies
Title and Abstract		
Title	1	Identify the study as an economic evaluation or use more specific terms such as "cost-effectiveness analysis," and describe the interventions compared.
Abstract	2	Provide a structured summary of objectives, perspective, setting, methods (including study design and inputs), results (including base case and uncertainty analyses), and conclusions.
Introduction		
Background and objectives	3	Provide an explicit statement of the broader context for the study. Present the study question and its relevance for health policy or practice decisions.
Methods		
Target population and subgroups	4	Describe characteristics of the base case population and subgroups analyzed, including why they were chosen.
Setting and location	5	State relevant aspects of the system(s) in which the decision(s) need(s) to be made.
Study perspective	6	Describe the perspective of the study and relate this to the costs being evaluated.
Comparators	7	Describe the interventions or strategies being compared and state why they were chosen.
Time horizon	8	State the time horizon(s) over which costs and consequences are being evaluated and say why they are appropriate.
Discount rate	9	Report the choice of discount rate(s) used for costs and outcomes and say why they are appropriate.
Choice of health outcomes	10	Describe what outcomes were used as the measure(s) of benefit in the evaluation and their relevance for the type of analysis performed.
Measurement of effectiveness	11a	Single study-based estimates: Describe fully the design features of the single effectiveness study and why the single study was a sufficient source of clinical effectiveness data.
	11b	Synthesis-based estimates: Describe fully the methods used for identification of included studies and synthesis of clinical effectiveness data.
Measurement and valuation of preference-based outcomes	12	If applicable, describe the population and methods used to elicit preferences for outcomes.
Estimating resources and costs	13a	Single study-based economic evaluation: Describe approaches used to estimate resource use associated with the alternative interventions. Describe primary or secondary research methods for valuing each resource item in terms of its unit cost. Describe any adjustments made to approximate to opportunity costs.
	13b	Model-based economic evaluation: Describe approaches and data sources used to estimate resource use associated with model health states. Describe primary or secondary research methods for valuing each resource item in terms of its unit cost. Describe any adjustments made to approximate to opportunity costs.
Currency, price, date, and conversion	14	Report the dates of the estimated resource quantities and unit costs. Describe methods for adjusting estimated unit costs to the year of reported costs if necessary. Describe methods for converting costs into a common currency base and the exchange rate.
Choice of model	15	Describe and give reasons for the specific type of decision-analytical model used. Providing a figure to show model structure is strongly recommended.

Continued

TABLE 19.7 Items Present in a Typical Pharmacoeconomic Analysis as per CHEERS Statement 2013[25]—cont'd

Section	Item No.	Recommendation to Authors of Pharmacoeconomic Studies
Assumptions	16	Describe all structural or other assumptions underpinning the decision-analytical model.
Analytical methods	17	Describe all analytical methods supporting the evaluation. This could include methods for dealing with skewed, missing, or censored data; extrapolation methods; methods for pooling data; approaches to validate or make adjustments (such as half cycle corrections) to a model; and methods for handling population heterogeneity and uncertainty.
Results		
Study parameters	18	Report the values, ranges, references, and, if used, probability distributions for all parameters. Report reasons or sources for distributions used to represent uncertainty where appropriate. Providing a table to show the input values are strongly recommended.
Incremental costs and outcomes	19	For each intervention, report mean values for the main categories of estimated costs and outcomes of interest, as well as mean differences between the comparator groups. If applicable, report incremental cost-effectiveness ratios.
Characterizing uncertainty	20a	*Single study-based economic evaluation*: Describe the effects of sampling uncertainty in the estimated incremental cost and incremental effectiveness parameters, together with the impact of methodological assumptions (such as discount rate, study perspective).
	20b	*Model-based economic evaluation*: Describe the effects on the results of uncertainty for all input parameters, and uncertainty related to the structure of the model and assumptions.
Characterizing heterogeneity	21	If applicable, report differences in costs, outcomes, or cost-effectiveness that can be explained by variations between subgroups of patients with different baseline characteristics or other observed variability in effects that are not reducible by more information.
Discussion		
Study findings, limitations, generalizability, and current knowledge	22	Summarize key study findings and describe how they support the conclusions reached. Discuss the limitations and the generalizability of the findings and how the findings fit with current knowledge.
Other		
Source of funding	23	Describe how the study was funded and the role of the funder in the identification, design, conduct, and reporting of the analysis. Describe other nonmonetary sources of support.
Conflicts of interest	24	Describe any potential for conflict of interest of study contributors in accordance with journal policy. In the absence of a journal policy, we recommend authors comply with International Committee of Medical Journal Editors recommendations.

statement, a typical pharmacoeconomic article shall have the following items. A small description of these items shall give a rough idea of the components present in a typical pharmacoeconomic analysis. The list of items in the statement provides an idea about different parameters found in a typical pharmacoeconomic study. The CHEERS statement focuses on the quality of reporting health economic studies, not the quality of how those studies were conducted.

For consistency, the CHEERS statement checklist format is based on the format of the CONSORT statement checklist.

The **ISPOR CHEERS Task Force Report** provides examples and further discussion of the 24-item CHEERS checklist and the CHEERS statement. It may be accessed via the *Value in Health* link or via the ISPOR Health Economic Evaluation Publication Guidelines—CHEERS: Good Reporting Practices web page.[25]

As the format of reporting guidelines like CHEERS covers key concepts in pharmacoeconomic studies, it familiarizes the reader to such literature.

CONCLUSION

Pharmacoeconomics is the scientific discipline that evaluates the value of drug or pharmacy services. Pharmacoeconomic analyses provide a measure called ICER that is commonly used to assist healthcare decision making. The correct and appropriate interpretations of pharmacoeconomic analysis findings are crucial for any level of healthcare decision making. Pharmacoeconomic reporting guidelines such as CHEERS direct the reports to be in a common format for easy reading of the pharmacoeconomic literature.